

Neurology Billing Solutions

Maximizing Revenue Across E/M, EEG/EMG, Infusion Therapy, Botox, and Neuromodulation.

Prepared by: Aethera Healthcare Solutions · Lakeland, FL

Scope: CPT 95800–96020 · 99202–99215 · 96365 · J0585 Botox ·

Updated: 95800–96020 · Aetherahealthcare.com · +1 (863) 694-0325



ABOUT AETHERA

Maximizing Revenue. Minimizing Burden.

Aethera Healthcare Solutions is your full-service medical billing partner headquartered in Lakeland, FL. We handle coding, claims, payments, denials, and collections — so you can focus on patients.

500+

Practices Served

Across 25+ medical specialties

\$12M+

Revenue Recovered

Denied & underbilled claims recouped

95%

Clean Claim Rate

vs 84% industry avg

< 30d

Average AR

Days in accounts receivable



Personal, Not Corporate

Dedicated account team with deep specialty and payer knowledge. You work with the same names every week — no rotating call-center handoffs.



Performance Targets, In Writing

96%+ net collection · < 5% denial · 95%+ coding accuracy · 24h posting · 48h submission · 65%+ appeal success.



Zero Risk To Start

No setup fees, no long-term contracts. 90-day cancel anytime. Free 5-day audit before any commitment.



NEUROLOGY · MARKET CONTEXT

Why Neurology Billing Matters Now

Neurology is a high-complexity specialty driven by aging demographics — Alzheimer's, Parkinson's, stroke, MS. Diagnostic testing (EEG, EMG, NCS) represents 30–40% of practice revenue and is the most-bundled category in neurology billing.



\$108B

US neurological disease treatment spend (2024)

Includes stroke, AD, PD, MS, epilepsy, migraine.



6.7M

Americans with Alzheimer's disease (2024)

Projected to reach 12.7M by 2050.



~ 19%

Neurology claim denial rate (industry avg)

Driven by modifier -26/-TC errors and missing pre-auth for infusion therapy.



\$24B+

Annual US MS disease-modifying therapy spend

Tysabri, Ocrevus, Tecfidera dominate at \$7K–\$9K/month.



Net effect: neurology practices that master modifier -26/-TC splits, Botox documentation, and infusion PA workflow can recover **8–14% of total revenue** currently lost to denials and under-billing.



NEUROLOGY · OVERVIEW

The Four Pillars of Neurology Billing

Each pillar represents a distinct service family with unique coding, modifier, and payer requirements.



01 · E/M + HEADACHE

Office E/M & Headache Management

Office visits (99202–99215) for chronic conditions: migraine, epilepsy, Parkinson's, MS, dementia. Botox for chronic migraine (J0585 + 64615).

CPT 99202–99215 · J0585 · 64615



02 · EEG/EMG/NCS

EEG, EMG & Nerve Conduction Studies

EEG (95800–95827), EMG (95860–95887), NCS (95900–95913), evoked potentials (95925–95938). Modifier -26/-TC split critical.

CPT 95800–96020 · Modifier -26/-TC



03 · INFUSIONS

MS & Neurology Infusions

Tysabri (J3359) for MS, Ocrevus (J9352) for MS, Soliris (J9246) for NMOSD, IVIG (J1568) for CIDP. Office infusion (96365).

J3359 · J9352 · 96365 · PA
mandatory



04 · NEUROMODULATION

DBS, VNS & Spinal Cord Stimulation

Deep brain stimulation programming (95983–95984), vagus nerve stimulator programming (95967), spinal cord stimulator programming (95972). Device management.

CPT 95983–95984 · 95967 · 95972



PILLAR 01 · E/M + HEADACHE

Office Visits & Chronic Migraine Botox

Neurology office encounters cover chronic conditions: migraine, epilepsy, Parkinson's, MS, dementia, neuropathy. The **2021 E/M coding reform** shifted documentation to MDM-based. Botox for chronic migraine (≥15 headache days/month) is FDA-approved and Medicare-covered with strict documentation. **Botox for cosmetic purposes is NEVER covered.**

1 E/M coding (2021+ reform)

99202–
99215

2 Botox for chronic migraine

J0585

64615

G43.709

3 CGRP inhibitors for migraine

J032

J033

J033



Documentation: Botox for chronic migraine requires ≥ 15 headache days/month, ≥ 4 hours/day, failed ≥ 2 preventive medications. Cosmetic Botox uses different diagnosis and is **NEVER billed to insurance.**



CPT Mini-Table · 2026 Avg \$

Top codes for office E/M and chronic migraine Botox administration.

Code	Service	Avg \$
99213	Est. E/M, low MDM	\$92
99214	Est. E/M, moderate MDM	\$131
99215	Est. E/M, high MDM	\$187
J0585	Botox, per unit (chronic migraine)	\$6
64615	Chemodenervation cranial nerve VII	\$425



PILLAR 02 · DIAGNOSTIC TESTING

EEG, EMG, NCS & Evoked Potentials

Diagnostic testing is the procedural backbone of neurology. Every test has a **professional component** (interpretation, modifier -26) and a **technical component** (equipment/technician, modifier -TC). Correct modifier use determines whether the claim pays at all.



EEG — Electroencephalography

95800 EEG without video, awake/drowsy (\$425) · 95806 EEG with sleep, full night (\$750) · 95810 video EEG, 2–12 hours (\$1,150) · 95819 video EEG, 12–26 hours (\$1,850).



EMG — Electromyography

95860 needle EMG, 1 extremity (\$185) · 95863 needle EMG, 3 extremities (\$375) · 95867 needle EMG, paraspinal muscles (\$125). Diagnostic for neuropathy, ALS, myopathy.



NCS — Nerve Conduction Studies

95900 NCS, motor, 1 nerve (\$85) · 95904 NCS, sensory, 1 nerve (\$85) · 95913 NCS, 11–12 nerves (\$525). Paired with EMG for full diagnostic workup.



Evoked Potentials

95925 VEP, visual evoked potential (\$325) · 95930 BAEP, brainstem auditory (\$295) · 95934 SSEP, somatosensory, upper (\$425). Used for MS, optic neuritis, hearing assessment.

-26 Modifier -26 (Professional)

Billed by interpreting neurologist. Covers interpretation + report only. Cannot bill without separate written report in chart.

-TC Modifier -TC (Technical)

Billed by facility/equipment owner. Covers technician time + equipment. Cannot bill -26 and -TC on the same claim.

⌘ Bundling Rules

95860 EMG + 95900 NCS same extremity, same session: separately billable. Multiple NCS nerves same session: bill per nerve (95900 × N). Same-day multiple EEG durations: only longest billable.



PILLAR 03 · INFUSIONS

MS Disease-Modifying Therapies & IVIG

Multiple sclerosis (MS) disease-modifying therapies are the highest-cost outpatient neurology drugs. **Tysabri** (natalizumab) costs ~ **\$7,000/infusion** every 4 weeks; **Ocrevus** (ocrelizumab) costs ~ **\$65,000/dose** every 6 months. IVIG for CIDP, MMN, and other neuropathies costs **\$5,000–\$10,000/infusion**. Pre-auth mandatory for all; step therapy documentation required.

1 Tysabri (J3359)

\$7,000/infusion
96365

2 Ocrevus (J9352)

**\$65,000/
dose**

3 IVIG (J1568)

\$0.85/gm

\$70/infusion

➤ MS infusion PA denial rate averages **25–35%** on first submission. Aethera pre-auth workflow with step therapy documentation reduces to **<10%**.

\$ KPI · Infusion Therapy Economics

\$7,000

Tysabri per infusion
J3359 · every 4 weeks maintenance

\$65,000

Ocrevus per dose
J9352 · 6-month interval

\$5K–\$10K

IVIG per infusion range
J1568 · for CIDP, MMN, neuropathies

\$24B+

Annual US MS therapy spend
Tysabri, Ocrevus, Tecfidera dominate

■ PILLAR 04 · NEUROMODULATION

DBS, VNS & Spinal Cord Stimulation Programming

Neuromodulation devices (deep brain stimulators, vagus nerve stimulators, spinal cord stimulators) require periodic programming to optimize therapy. Each programming session is billable; quarterly cycles create recurring revenue per patient.



Deep Brain Stimulation (DBS)

95983 DBS programming, first 30 min (\$165); 95984 each addl 15 min (\$75). Used for Parkinson's, essential tremor, dystonia. Billed quarterly.



Vagus Nerve Stimulator (VNS)

95967 VNS programming, single programming session (\$115). Used for refractory epilepsy and treatment-resistant depression. Quarterly cycle.



Spinal Cord Stimulation (SCS)

95972 SCS programming, single session (\$135); 95973 complex programming (\$185). Used for chronic pain, failed back surgery syndrome.



Device Implantation & Revision

61863 DBS implantation, single array (\$3,200); 64568 VNS im plantation (\$1,800); 63650 SCS percutaneous implantation (\$2,400). Surgical codes; pre-auth mandatory.

PER-SESSION REIMBURSEMENT

\$165

Per DBS programming session
(CPT 95983)

\$115

Per VNS programming session
(CPT 95967)

\$135–\$185

Per SCS programming session range

~ \$660/yr

Annual recurring revenue per DBS patient
(quarterly)



Compliance alert: Device programming requires documentation of parameter changes (amplitude, pulse width, frequency, electrode configuration). Missing parameter documentation is the #1 neuromod denial.



NEUROLOGY · BILLING CODES

Top CPT Codes & 2026 National Average Reimbursement

Most-billed codes across the four neurology pillars. Professional component shown (-26 modifier).

Code	Service	Modifier	Avg \$	Notes
99213	Est. E/M, low MDM	—	\$92	Most common neuro E/M
99214	Est. E/M, moderate MDM	—	\$131	MS/epilepsy management
99215	Est. E/M, high MDM	—	\$187	Complex neurological case
J0585	Botox per unit (chronic migraine)	—	\$6	155 units per treatment
64615	Chemodenervation cranial nerve VII	—	\$425	Botox admin code
95806	EEG with sleep, full night	-26 or -TC	\$750	Sleep study / seizure workup
95860	Needle EMG, 1 extremity	-26 or -TC	\$185	Neuropathy diagnosis
95900	NCS, motor, 1 nerve	-26 or -TC	\$85	Pair with EMG
95913	NCS, 11-12 nerves	-26 or -TC	\$525	Comprehensive NCS
95810	Video EEG, 2-12 hours	-26 or -TC	\$1,150	Epilepsy monitoring
J3359	Tysabri (natalizumab), per 1mg	—	\$7	~\$7K per infusion

MODIFIER REFERENCE

— 26 — professional component (interpretation only). —TC — technical component (equipment/technician). Cannot bill both same claim. 25 — significant, separately identifiable E/M same day as procedure.

BOTOX DOCUMENTATION

J0585 for chronic migraine: diagnosis G43.709 required, documented ≥15 headache days/month, ≥2 failed preventives. Cosmetic Botox: NEVER billed to insurance.

NEUROLOGY · DENIAL PREVENTION

Top 8 Neurology Denial Reasons & How Aethera Prevents Each

Neurology denial rates average 17–21%; each denied claim costs \$50–150 to rework. Diagnostic testing modifier errors and infusion PA denials drive 50%+ of revenue loss.

!RED 24 % Modifier – 26/– TC errors CAUSE:Diagnostic test billed without proper modifier; either double-billed (both –26 and –TC) or modifier missing entirely. Aethera modifier rules engine enforces correct –26/–TC per provider/facility setup.	!RED 19 % MS infusion PA missing CAUSE:Tysabri/Ocrevus infusion administered without prior authorization on file; \$7K–\$65K denied. Aethera PA tracker auto-submits before scheduling; verifies approval before infusion.
!AMBER 14 % Botox documentation insufficient CAUSE:J0585 + 64615 billed without headache diary, failed preventive documentation, or G43.709 diagnosis. Aethera Botox documentation checklist before claim release.	!AMBER 11 % Cosmetic Botox billed as medical CAUSE:J0585 billed with cosmetic diagnosis; automatic denial + fraud risk. Aethera diagnosis verification blocks cosmetic Botox claims.
!AMBER 9 % Modifier 25 documentation insufficient CAUSE:E/M + diagnostic test same day without separate documentation supporting E/M significance. Aethera modifier 25 documentation checklist.	!GREEN 7 % DBS/VNS/SCS programming without parameter docs CAUSE:95983/95967/95972 billed without documented parameter changes. Aethera programming parameter documentation requirement.
!GREEN 6 % NCS overbilling CAUSE:Multiple NCS codes billed same session exceeding CMS-allowed nerve count per code. Aethera NCS bundling rules enforce CMS nerve-count limits.	!GREEN 5 % IVIG without step therapy documentation CAUSE:J1568 IVIG PA denied because prior steroid trial not documented. Aethera step-therapy documentation checklist before PA submission.
➔ Cumulative impact: Aethera reduces neurology denial rate from industry avg 19 % to < 7 % — recovering ~\$200K/year per \$1.5M practice revenue.	

■ CASE STUDY · INTEGRATED NEURO RCM

4-Neurologist Practice — Annual Revenue Impact

A \$2.9M-revenue practice models the combined upside of modifier -26/-TC discipline, MS infusion PA recovery, and Botox documentation capture.



STREAM 01 Modifier - 26/- TC Discipline

Annual diagnostic tests (EEG+EMG+NCS)	2,000
Pre-Aethera modifier error rate	22%
Post-Aethera error rate	5%
Recovered claims	340/yr
Avg diagnostic test value	\$325

ANNUAL REVENUE
\$110,500



STREAM 02 MS Infusion PA Recovery

MS biologic patient panel	60
Pre-Aethera PA denial rate	30%
Post-Aethera denial rate	9%
Recovered infusions	75/yr
Avg infusion value	\$3,800

ANNUAL REVENUE
\$285,000



STREAM 03 Botox + Programming Capture

Chronic migraine Botox patients	240
Pre-Aethera doc denial rate	18%
Post-Aethera denial rate	4%
Recovered treatments	168/yr
Avg Botox treatment value	\$1,500
DBS/VNS programming uplift	\$84,000

ANNUAL REVENUE
\$252,000



Combined annual revenue recovery: **320%** · Implementation cost (year 1): **\$100,000** (Aethera RCM fees + EHR integration + coder training) · Year-1 net: **\$100,000** **ROI**



Where OIG, Medicare RAC, and Commercial Payers Audit First

Top compliance risks across the four neurology pillars. Quarterly self-audits recommended.

 PILLAR 1 E/M + Headache	 PILLAR 2 Diagnostic Testing	 PILLAR 3 Infusions	 PILLAR 4 Neuromodulation
RED Cosmetic Botox as medical Billing J0585 with cosmetic diagnosis to insurance; civil fraud exposure under False Claims Act.	RED Modifier -26/-TC double-billing Billing both professional AND technical components on same claim; CMS recovers 2x reimbursement.	HIGH MS biologic without PA Administering \$7K-\$65K Tysabri/Ocrevus without prior auth; full denial + clawback risk.	RED Programming without parameter docs Billing 95983/95967 without documented amplitude, pulse width, frequency changes.
AMBER Botox documentation insufficient Billing J0585 + 64615 without headache diary, failed preventive documentation, or G43.709 diagnosis.	AMBER NCS overbilling Billing multiple 95900/95904 codes for nerve count exceeding CMS-allowed maximum per code.	MEDIUM Step therapy non-compliance Biologic PA denied because prior immunomodulator/steroid trials not documented.	AMBER Device implantation without PA 61863/64568/63650 surgical codes without pre-auth; full denial risk.
GREEN Modifier 25 misuse Appending to E/M when diagnostic test on same day does not represent significant, separately identifiable service.	LOW Same-session EEG overbilling Billing multiple EEG durations same session; only longest billable.	LOW IVIG without CIDP diagnosis PA denied without nerve conduction study supporting CIDP for J1568.	LOW Programming frequency over-limit Billing DBS programming more than quarterly without medical necessity documentation.



OIG Work Plan 2025-2026 explicitly lists modifier -26/-TC errors in neurology diagnostic testing and Botox documentation as active audit targets. Medicare RACs have recovered **> \$180M** from neurology practices since 2020.



Three Actions to Capture Neurology Revenue

For neurology practice administrators and revenue cycle directors leading the next 90 days.

01

Deploy modifier –26/–TC rules engine

Audit last **200 diagnostic test claims** for modifier –26/–TC appropriateness. Implement pre-submission rules engine that blocks claims with missing or duplicated modifiers. Target: **< 5 %** modifier error rate (industry avg **22 %**).

↓ **5 % modifier error rate**

02

Automate MS infusion PA workflow

Build EHR template that auto-submits PA at **Tysabri/Ocrevus** scheduling. Document step therapy (immunomodulator/steroid failures) before first PA submission. Pre-auth denial rate drops from **30 %** to **< 9 %** — recovering ~ **\$285 K** per 60-patient MS panel.

↑ **\$285 K per MS panel**

03

Enforce Botox + programming documentation

Build EHR template requiring headache diary, failed preventive documentation, and **G43.709** diagnosis before Botox claim release. Implement parameter documentation requirement for DBS/VNS/SCS programming (95983/95967/95972). Recovers ~ **\$336 K/year** for typical practice.

↑ **\$336 K/year recovery**

Questions & Discussion

Optimizing Neurology Revenue Across E/M, Diagnostics, Infusions & Neuromodulation

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